Project Design Phase-I
Solution Architecture

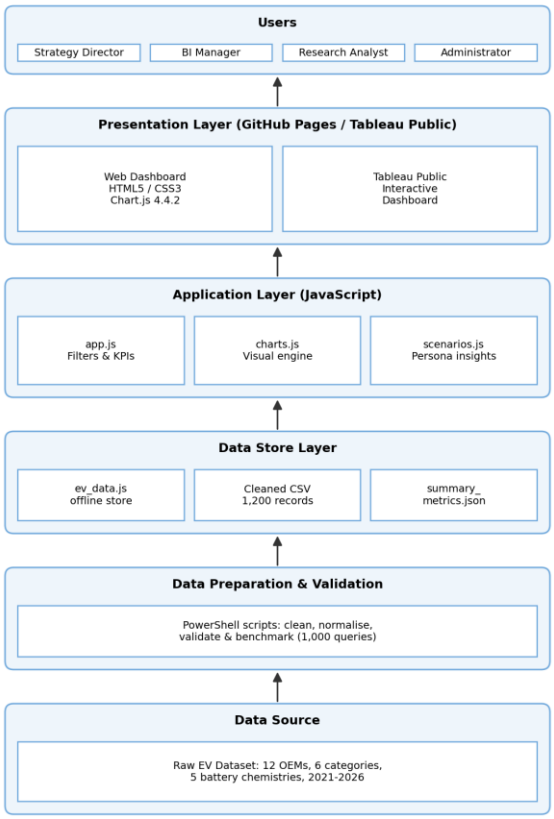
Date	23 October 2026
Team ID	0157CY241012
Project Name	EV Industry Analytics & Intelligence Platform
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Source	Synthetic EV dataset: 1,200 records, 20 columns, 12 OEMs, 6 categories, 5 battery chemistries (2021-2026).	CSV, PowerShell (generate_data.ps1)
Data Preparation & Validation	Cleans nulls, normalises data types, calculates price-per-km and CO2 metrics, runs integrity and range checks.	PowerShell, CSV, JSON
Data Store	Holds cleaned dataset, precomputed summary metrics and the offline data bridge used by the web app.	ev_market_data_cleaned.csv, summary_metrics.json, ev_data.js
Application Logic	Manages filters, KPI counters and persona scenario insights (Emily, David, Dr. Sarah).	JavaScript (app.js, scenarios.js)
Visualization Layer	Renders 10+ charts and a multi-tab responsive dashboard with dark/light themes; Tableau dashboard for analysis.	Chart.js 4.4.2, HTML5, CSS3, Tableau Public
Hosting & Testing	Publishes the live demo and benchmarks query performance (1,000 queries in 7.6 ms).	GitHub, GitHub Pages, PowerShell (test_performance.ps1)